SPACE FACULTY

Let's learn about **satellites**!

Space mission and **satellite** design course

Launch systems and space logistic - Interplanetary Missions

2NDSpace

What are we going to talk about today?

LAUNCH SYSTEMS

TRANSFER VEHICLES

THE FUTURE OF SPACE LOGISTIC

PROPULSION TECHNOLOGY FOR LAUNCHERS

INTERPLANETARY MISSIONS

LAUNCH SYSTEMS

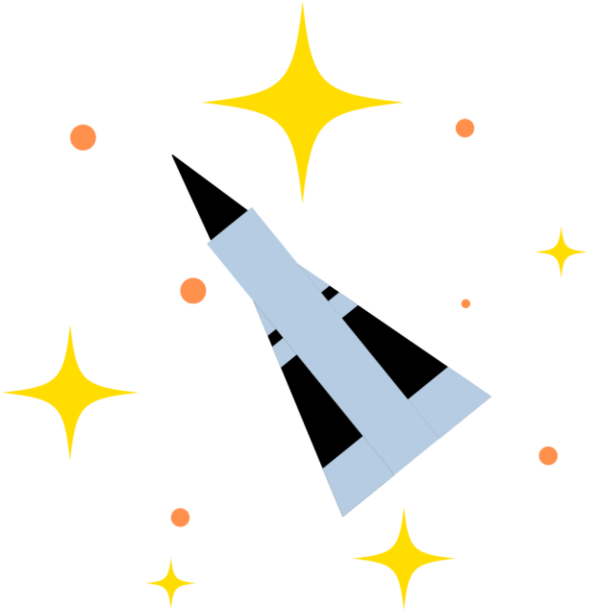
What is a space launcher?

It's a particular **rocket** designed to **carry a payload** into space.

The payload could be a satellite or human crew.

It can reach a low Earth orbit as the one of the International Space Station to carry astronauts and materials, or it can go to geostationary orbits to carry telecommunication satellites.

Special launcher are design to reach also the Moon or other planets.



Types of space launchers

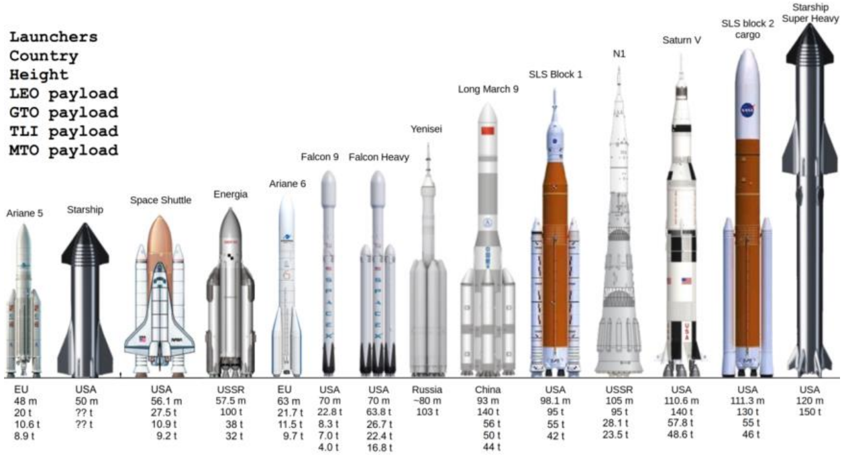
Launch vehicles are classed according to **low Earth orbit payload capability**:

★ **Small-lift launch vehicle:**
< 2,000 kg - e.g. Vega

★ **Medium-lift launch vehicle:**
2,000 to 20,000 kg - e.g. Soyuz ST

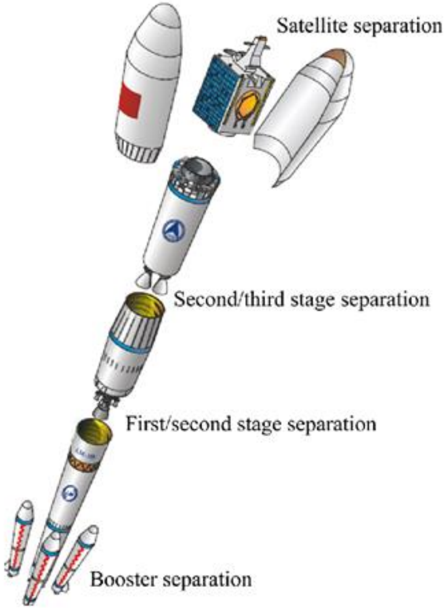
★ **Heavy-lift launch vehicle:**
> 20,000 to 50,000 kg - e.g. Ariane 5[5]

★ **Super-heavy lift vehicle:**
> 50,000 kg- e.g. Saturn V



Types of space launchers

In addition, launch vehicles also differ based on the number of stages they are composed of and the possibility of reuse.



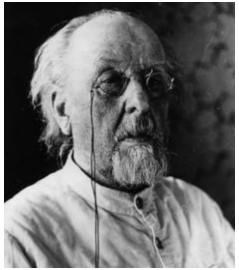
History of space launchers

Archytas

428-347 B.C.

A.D. 1232

Chinese Fire
Arrows

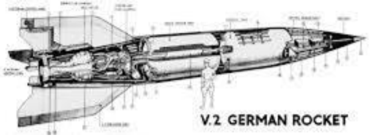


Tsiolkovsky

1903

1926

Goddard



Oberth -
V2 rocket

1944

History of space launchers

R-7
(Sputnik)

1957

1958

Juno 1
(Explorer 1)

Atlas
(Friendship 7)

1962

1969

Saturn V
(Apollo 11)

Titan
(Voyager)

1977

2NDSpace

History of space launchers

Space Shuttle

1981

1990

Pegasus

New Shepard

2015

2015

Falcon 9 (Full thrust)

Space Launch System

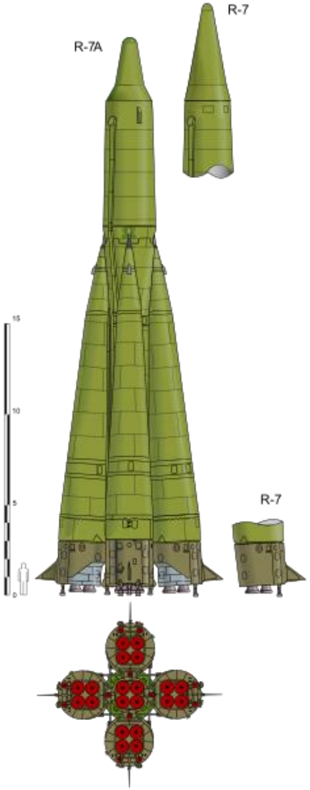
2022

R-7 Family

The R-7 Semyorka was a Soviet missile developed during the Cold War and was born as an **intercontinental ballistic missile**.

- ★ **34 m** high;
- ★ **10.3 m** in diameter;
- ★ weighed **280** metric **tons**.

It's first version had a **single stage** with **four strap on boosters** powered by rocket engines using liquid oxygen (**LOX**) and **kerosene** and capable of delivering its payload up to 8,000 km.

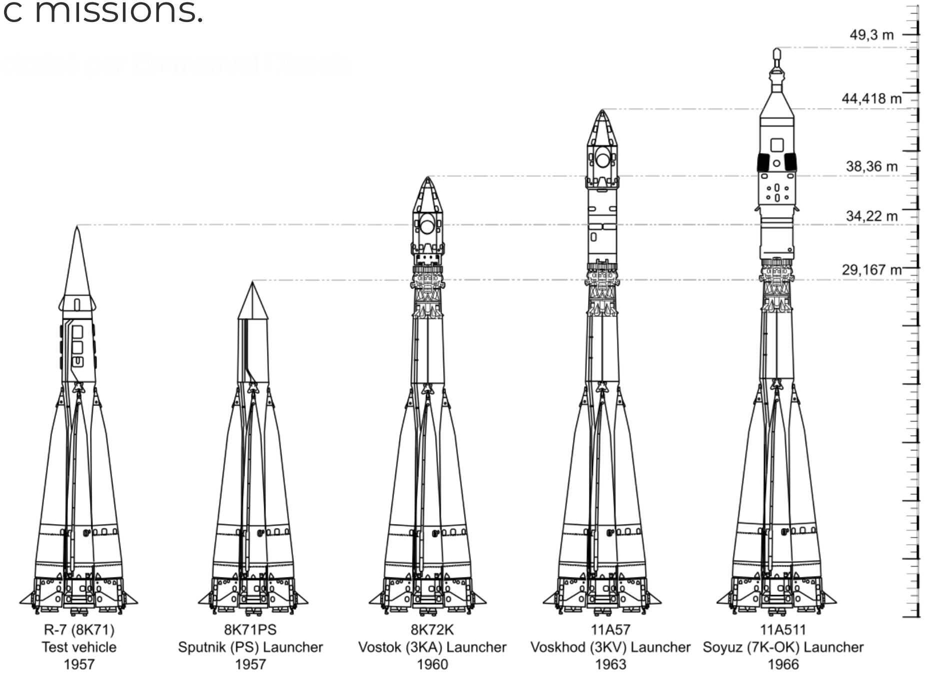


R-7 Family

There have been a number of variants of the R-7 with an upper stage and each optimized to carry out specific missions.

An R-7 was used to launch **Sputnik 1** (October 4, 1957) and an R-7 variant, the **Vostok**, launched the cosmonaut **Jurij Gagarin** (April 12, 1961), who became the first human to orbit Earth.

A multipurpose variant, the **Soyuz**, was first used in 1966 and, with many subsequent variants and improvements, it is **still in service**.



Saturn V

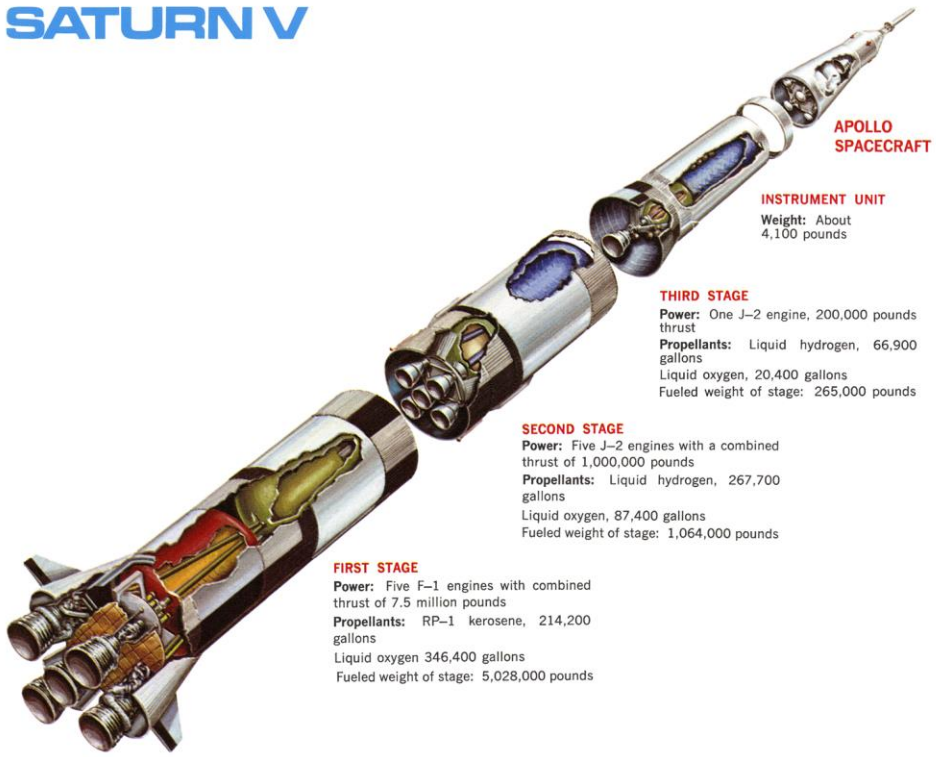
The Saturn V was an American super heavy-lift launch vehicle developed by NASA under the Apollo program for human exploration of the Moon.

It flown from **1967 to 1973**, it was used for nine crewed flights to the Moon, and to launch Skylab, the first American space station.

Saturn V

The Saturn V is a 3-stage rocket.

★ **First Stage** used RP-1 (a specially densified kerosene) and liquid oxygen (LOX) as the propellant to power its five rocket engines
★ **Second & Third stages** used cryogenic liquid hydrogen (LH) and LOX.



The Saturn V was **85% propellant by mass** on the launch pad. The first stage of this rocket produced over 3,000 tons of thrust at liftoff.

Space Shuttle (STS)

The Space Shuttle was a partially reusable low Earth orbital spacecraft system operated from **1981 to 2011** by NASA.

Operational missions launched numerous satellites, interplanetary probes, and the **Hubble Space Telescope** (HST), conducted science experiments in orbit and participated in the construction and servicing of the International Space Station (**ISS**).

Four fully operational orbiters were initially built, but two were lost in mission accidents: Challenger in 1986 and Columbia in 2003, with a total of 14 astronauts killed.

Space Shuttle (STS)

The STS was composed of:

- ★ **Orbiter Vehicle**

  with 3 liquid-fuel cryogenic rocket engines. On returning to Earth, it re-entered the atmosphere and landed like a supersonic glider on a runway.

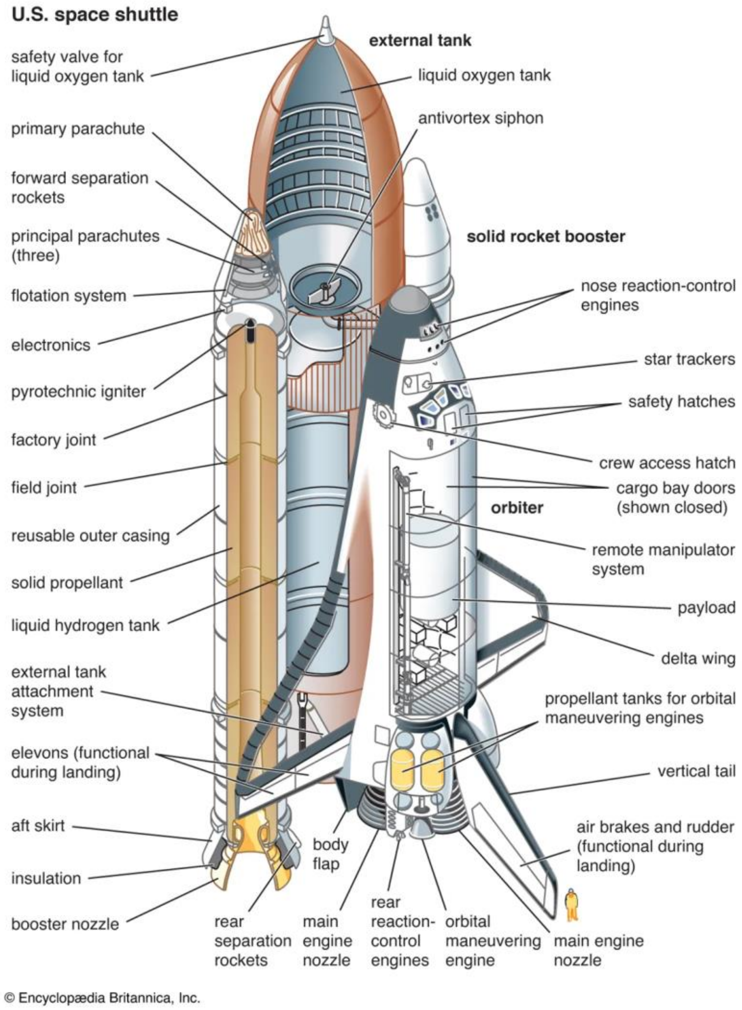
  

- ★ **External Tank** containing liquid hydrogen and liquid oxygen.

- ★ **Solid Rocket Boosters** with ammonium perchlorate (oxidizer) and atomized aluminum powder (fuel). Jettisoned after about 90s into the flight, parachuted into the sea, recovered, and reused.

Space Shuttle (STS)



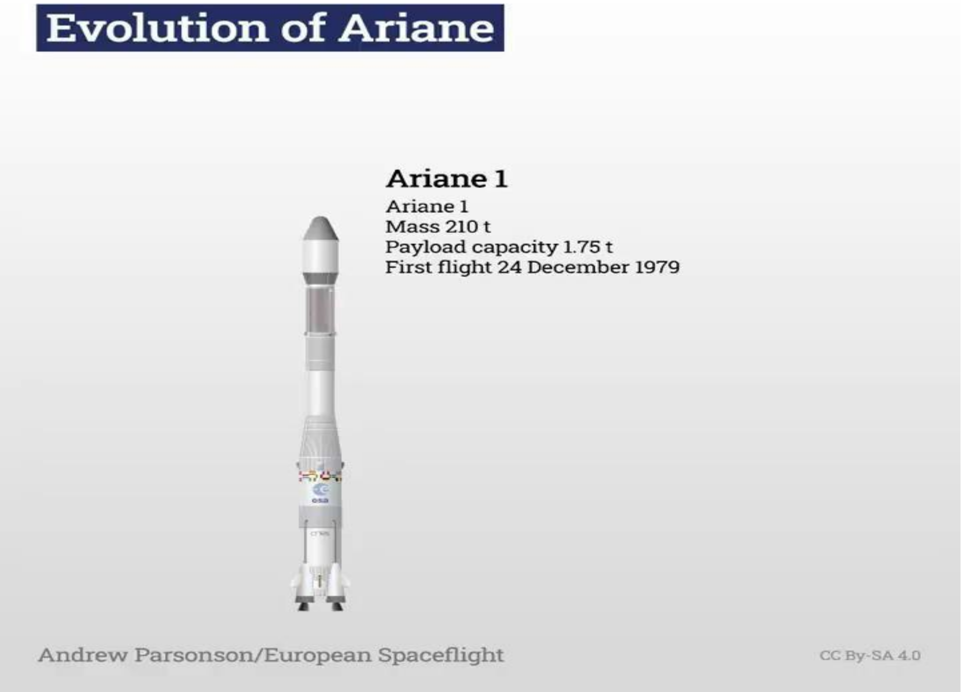
ESA’s launchers

ESA’s launchers - Ariane Family

Ariane 1 was the first ESA expendable launch system.

It was the first launcher to be developed with the primary purpose of **sending commercial satellites into geosynchronous orbit**. As the size of satellites grew, Ariane 1 quickly gave way to the more powerful Ariane 2, Ariane 3 and Ariane 4, heavily based upon the original rocket.

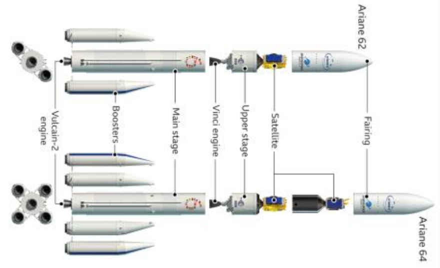
The successor rocket Ariane 5 uses a far greater proportion of all-new elements.



ESA’s launchers - Ariane 5 and 6

Ariane 5 (1996-2023) is a two stage launcher made of a cryogenic main stage, with two compartments, one for liquid oxygen and one for liquid hydrogen, and a Vulcain 2 rocket engine, two solid rocket boosters and a second stage positioned on top of the main stage and below the payload.

Ariane 6 is the launch system that replaces the Ariane 5, as part of the Ariane launch vehicle family. The stated motivation for Ariane 6 was to halve the cost compared to Ariane 5, and increase the capacity for the number of launches per year (from six or seven to up to eleven).



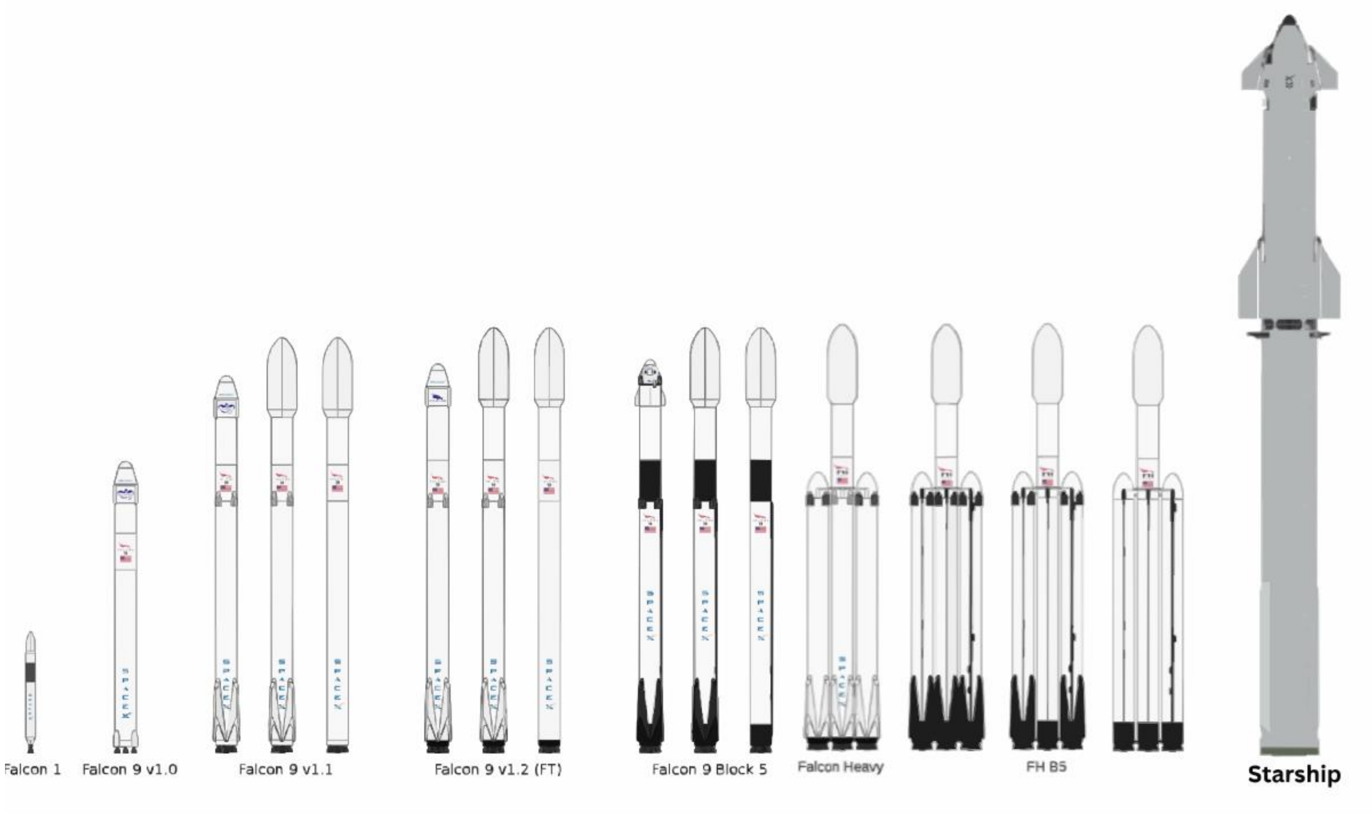
ESA’s launchers - Vega Family

Vega’s first launch took place in **2012** and represented a new-generation vehicle for flights with small to medium-sized satellite payloads.

Unlike most small rockets, Vega is able to lift payloads ranging from a single satellite up to one main satellite plus additional small satellites and **place them into separate orbits on a single mission**. This has made reaching space cheaper, quicker and easier.

It has **three solid-propellant stages** and **a liquid-propellant upper module** for attitude and orbit control as well as satellite release.

SpaceX launch vehicles



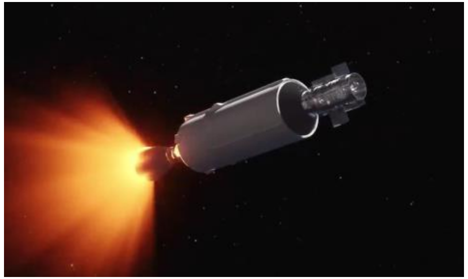
2NDSpace

SpaceX launch vehicles - Falcon 9

Falcon 9 is a reusable, two-stage rocket designed and manufactured for the reliable and safe transport of people and payloads into Earth orbit and beyond.

- ★ **First stage** incorporates 9 Merlin engines and aluminum-lithium alloy tanks containing liquid oxygen and rocket-grade kerosene (RP-1) propellant.

- ★ **Second stage** is powered by a single Merlin Vacuum Engine, delivers Falcon 9's payload to the desired orbit. It ignites a few seconds after stage separation, and can be restarted multiple times to place multiple payloads into different orbits.



SpaceX launch vehicles - Falcon 9 first stage

SpaceX launch vehicles - Starship

Made of:

- ★ **Starship spacecraft**
- ★ **Super Heavy rocket**
  powered by 33 Raptor engines using sub-cooled liquid methane (CH_4) and liquid oxygen (LOX),

  Represent a fully reusable transportation system designed to carry both crew and cargo to Earth orbit, the Moon, Mars and beyond. It is the world's most powerful launch vehicle ever developed.

Space Launch System (SLS)

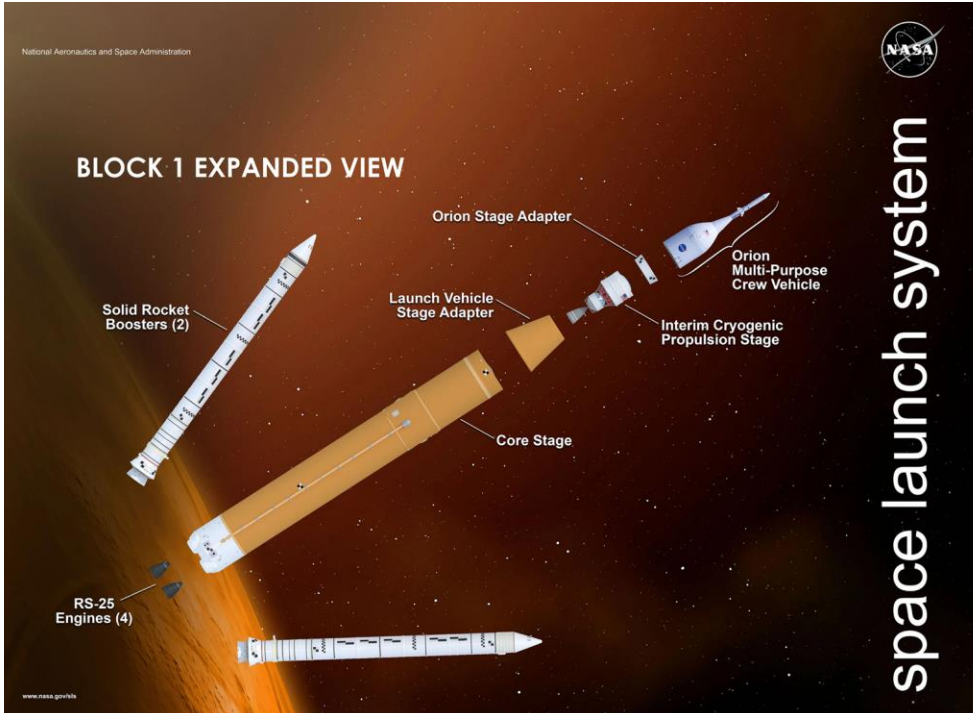
The SLS is an American **super heavy-lift expendable launch vehicle** used by NASA. As the primary launch vehicle of the Artemis Moon landing program, SLS is designed to launch the crewed Orion spacecraft on a trans-lunar trajectory. The first SLS launch was the uncrewed **Artemis 1**, which took place on 16 November **2022**.

Space Launch System (SLS) - stages

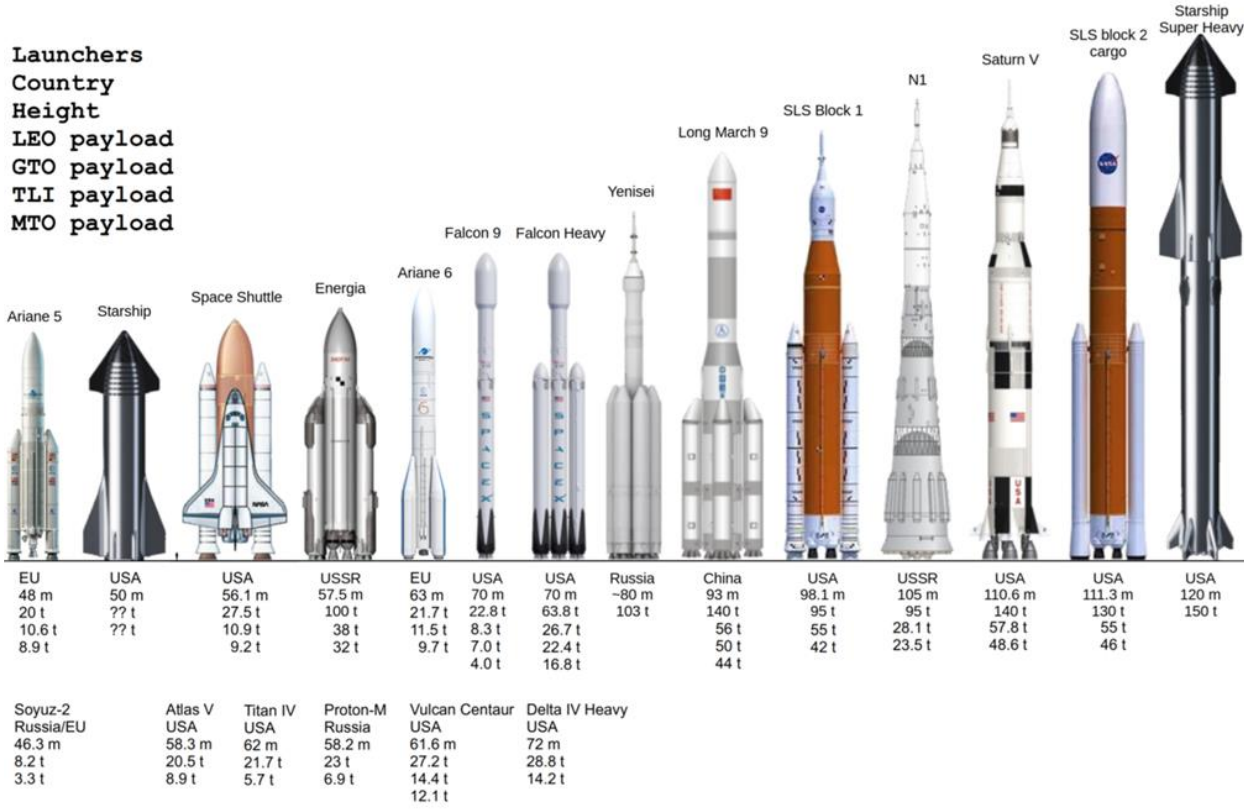
- ★ **Core stage**
  contains the LH and LOX tanks for the ascent phase and an assembly of the four RS-25 engines.

- ★ **Solid rocket boosters**

- ★ **Upper stage**
  cryogenic rocket stage consisting of a cylindrical LH tank structurally separated from an oblate spheroid LOX tank.



Comparison


Launchers
Country
Height
LEO payload
GTO payload
TLI payload
MTO payload
Ariane 5
Starship
Space Shuttle
Energia
Ariane 6
Falcon 9
Falcon Heavy
Yenisei
Long March 9
SLS Block 1
N1
Saturn V
SLS block 2 cargo
Starship Super Heavy
EU 48 m 20 t 10.6 t 8.9 t
USA 50 m ?? t ?? t
USA 56.1 m 27.5 t 10.9 t 9.2 t
USSR 57.5 m 100 t 38 t 32 t
EU 63 m 21.7 t 11.5 t 9.7 t
USA 70 m 22.8 t 8.3 t 7.0 t 4.0 t
USA 70 m 63.8 t 26.7 t 22.4 t 16.8 t
Russia ~80 m 103 t
China 93 m 140 t 56 t 50 t 44 t
USA 98.1 m 95 t 55 t 42 t
USSR 105 m 95 t 28.1 t 23.5 t
USA 110.6 m 140 t 57.8 t 48.6 t
USA 111.3 m 130 t 55 t 46 t
USA 120 m 150 t
Soyuz-2 Russia/EU 46.3 m 8.2 t 3.3 t
Atlas V USA 58.3 m 20.5 t 8.9 t
Titan IV USA 62 m 21.7 t 5.7 t
Proton-M Russia 58.2 m 23 t 6.9 t
Vulcan Centaur USA 61.6 m 27.2 t 14.4 t 12.1 t
Delta IV Heavy USA 72 m 28.8 t 14.2 t

PROPULSION TECHNOLOGIES FOR LAUNCHERS

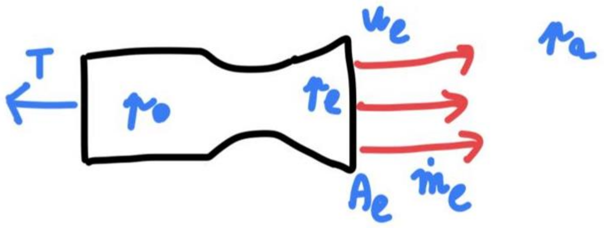
SPACE FACULTY

Thrust

It's the force that lift the rocket to the space.

This force is generated inside the combustion chamber by a propellent that produce heat and pressure.

The hot material is ejected through the nozzle and the resultant reaction move the rocket.



$$F = m * a = \dot{m}_e * u_e + (p_e - p_a) * A_e$$

$\dot{m}_e = mass\ flow\ rate$
$u_e = exhaust\ velocity$
$p_e = exhaust\ pressure$
$p_a = ambient\ pressure$
$p_0 = combustion\ chamber\ pressure$
$A_e = nozzle\ section\ area$

2NDSpace

Thrust

The total impulse generated by the rocket is equal to the total mass of propellent ejected multiplied to the equivalent exhaust velocity.

$$\boxed{I_{tot} = M_p * u_{eq}}$$

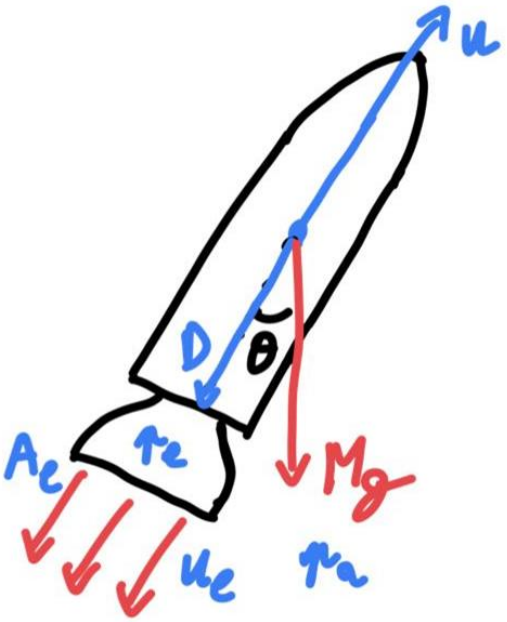
If divided by the propellant mass, we obtain the specific impulse characteristic of the engine.

$$\boxed{I_{sp} = u_{eq} \ \text{[m/s]}}$$

In this way is possible to compare different engine systems.

Speed Increment

The speed variation of the spacecraft is dependent on the **total mass ejected** from the nozzle, the **aerodynamics drag** and the **gravity effect**.



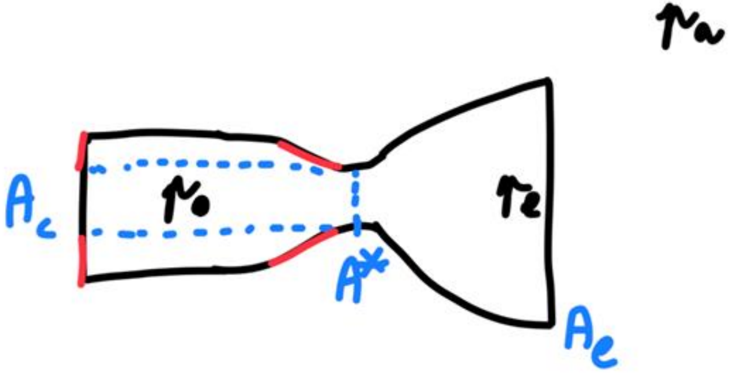
Where is the thrust applied?

After a force balance between all the pressure applied on the engine surface the resultant force is in the bottom of the engine

$$F = C_F * p_0 * A^*$$

C_F= thrust coefficient
A^*= throw section

A_e/A^* = expansion ratio



Mass efficiency

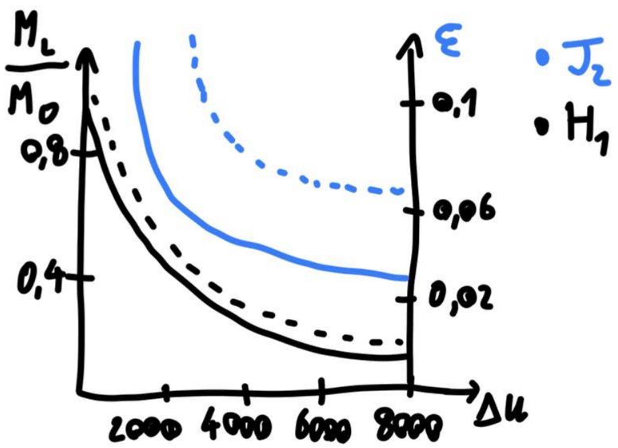
The total mass of the rocket is the sum of the propellant mass, the structure mass and the payload mass.

The structure mass is the sum of the engine mass and the propellant tank mass.

$$\boxed{M_0 = M_p + M_{eng} + M_{tank} + M_L}$$

$$\lambda = \frac{\mathrm{M_L}}{\mathrm{M_p} + \mathrm{M_s}}$$ ***Payload ratio***

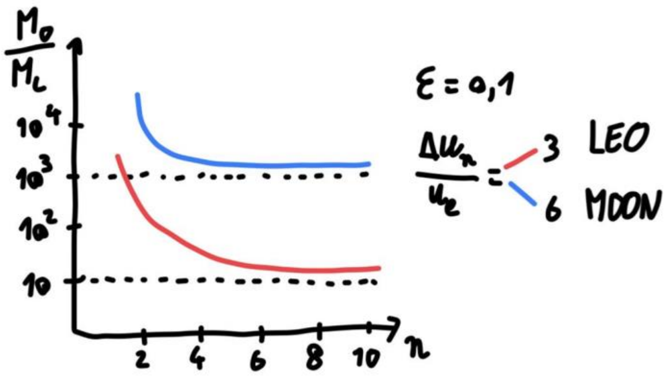
$$\varepsilon = \frac{\mathrm{M_s}}{\mathrm{M_0} - \mathrm{M_L}}$$ ***Structure ratio***



Multistage rockets

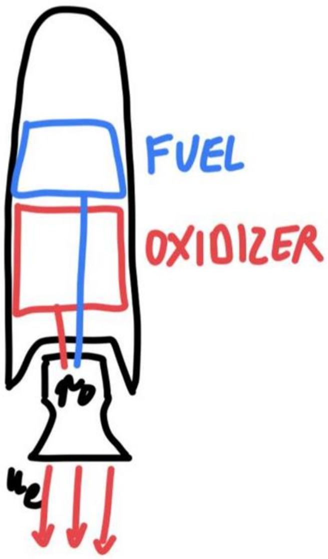
To reduce the structure mass it is possible to split the rocket in more parts and perform separation during the launch.

The total velocity increment is the sum of all the stage velocity and the payload ratio is function of the number of stages.



Chemical rockets

★ **Liquid propellant rocket**
fuel and oxidizer are separated in 2 tank. The oxidizer tank is closer to the combustion chamber to reduce the pipe length because the fluid is very reactive.

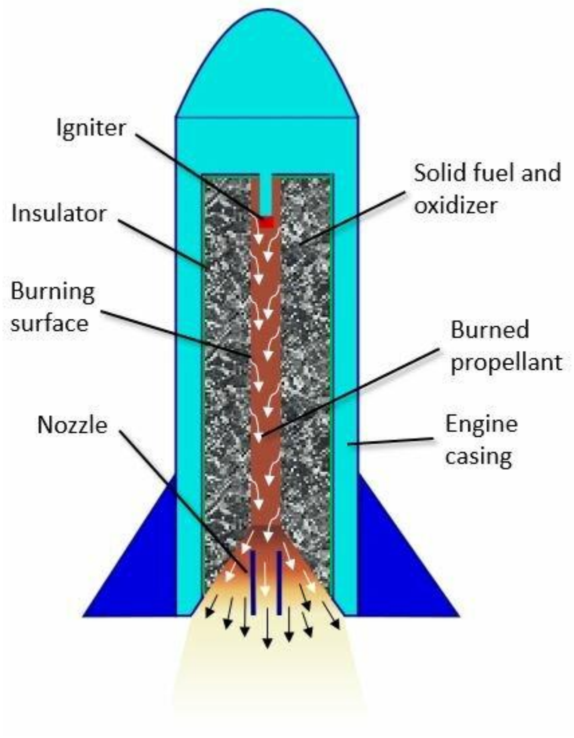


★ **Solid rocket booster**
the propellant fills the entire vessel and the combustion take place inside.

Solid rocket booster

- ★ **Advantages**
  - easy to store and handle;
  - high density and so small tanks to store them are required;
  - lower costs.

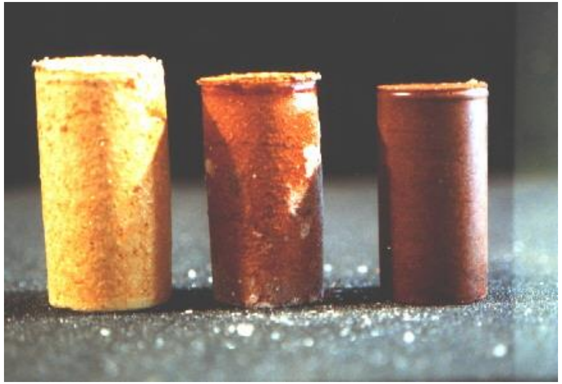
- ★ **Disadvantages**
  - specific impulse is low;
  - once it has been lit it will carry on burning until it is used up.



Types of solid propellants

★ **Double base:**
DB propellants are implemented in applications where minimal smoke is required

★ **Composite propellant:**
A powdered oxidizer and powdered metal fuel are intimately mixed and immobilized with a rubbery binder.



Preparation procedure

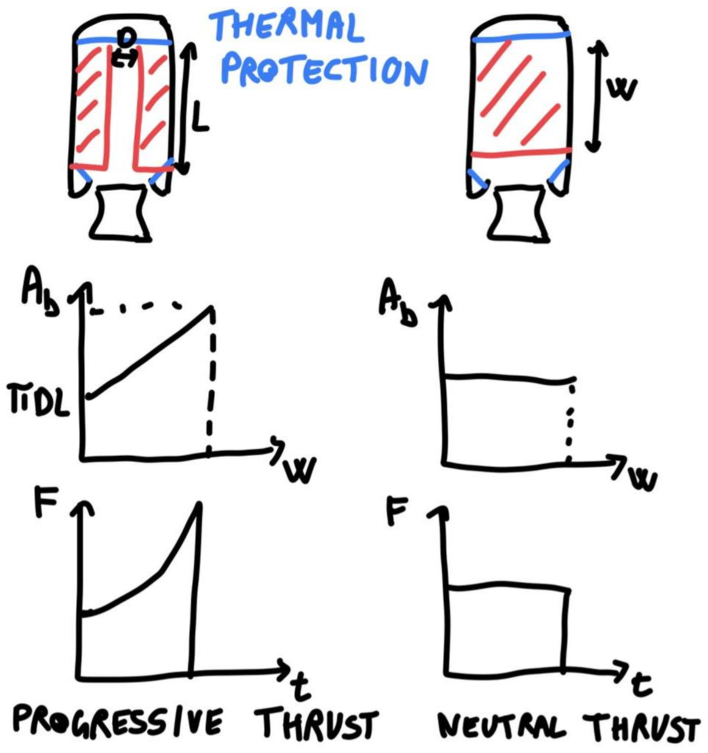
- ★ The oxidizer and the fuel powder are grinded in special mill to obtain the desired grain dimension.
- ★ The powders are mixed with binder and poured inside the tank.
- ★ Before the vulcanization process that make everything solid, all the air bubble are removed with vacuum and shaker system.
- ★ After the solidification, a complete TAC is performed to control imperfection inside the propellant.

Propellant geometry

The initial surface and mass distribution of the propellant inside the tank influences the burning rate.

For a launch system is preferable a progressive or reverse thrust. The presence of propellant over the tank until the burnout time also protect from the extreme heat the structure.

In a missile a neutral thrust is preferred to reach the maximum thrust, even if the structure is exposed to heat, due to the short life of the rocket.



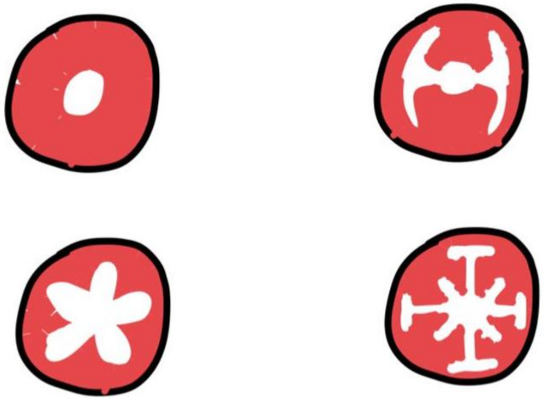
Propellant geometry

In main space launcher with payload is important to control the acceleration and structural stress.

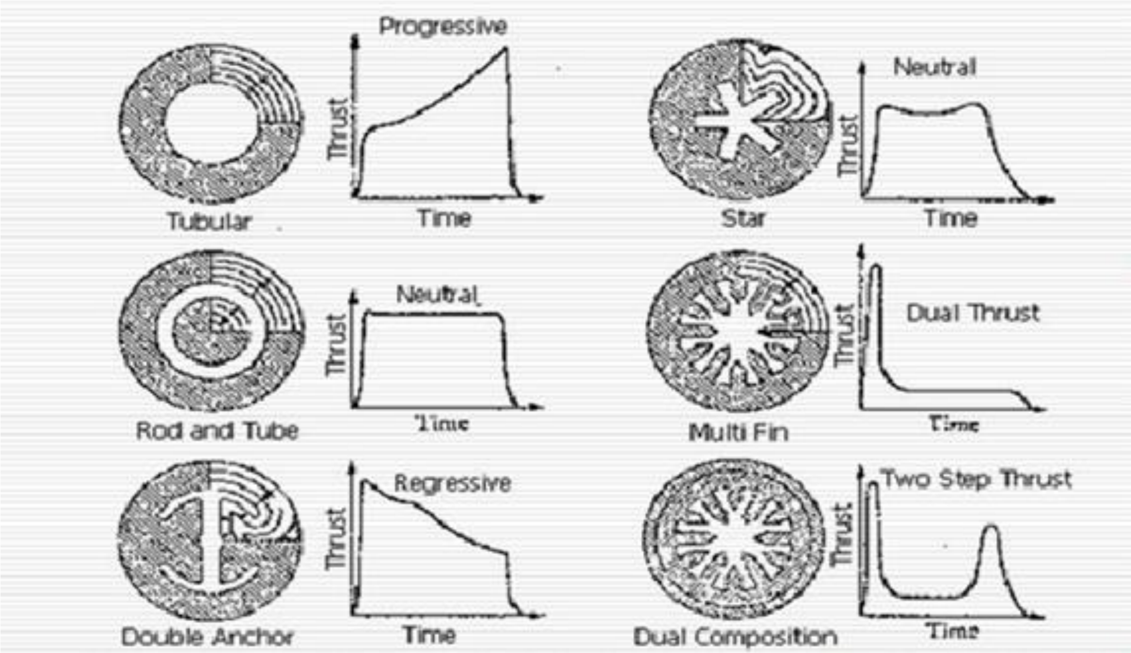
At the firing time it is needed a higher thrust due to the large mass.

However, as the material is expelled the total mass decreases.

So, to maintain the acceleration constant it is possible to reduce the burning rate.



Propellant geometry



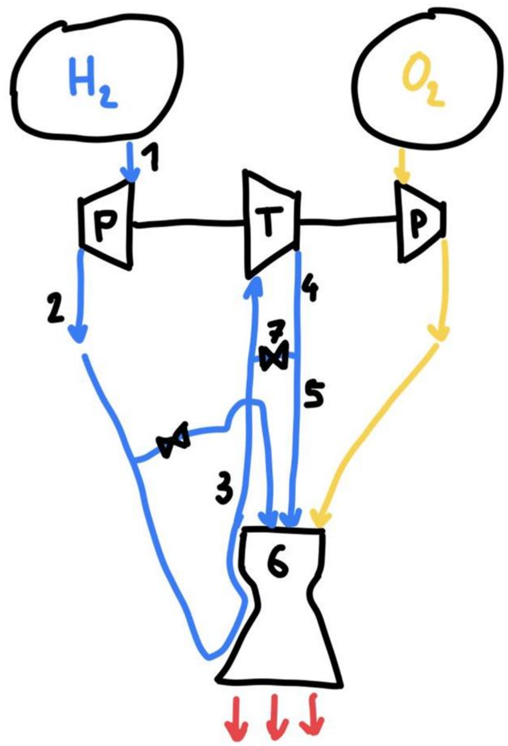
2NDSpace

SLS solid rocket booster static test

The purpose of this test is to ensure that the engine can produce the required thrust and operate correctly.

Liquid propellants

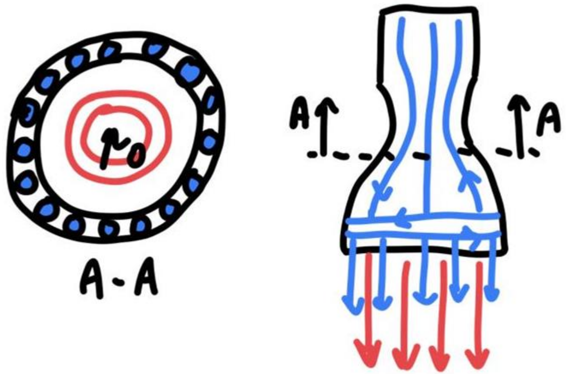
- **Advantages**
  - thrust can be controlled and engines can even be shut down and re-started at a later stage;
  - specific impulse is very large.
- **Disadvantages**
  - need for pumps, piping and separate storage for the fuel and oxidant means that extra mass has to be carried by the launch vehicle;
  - low density and so large tanks to store them are required.



Dump cooling

One of the main advantages of liquid propellants is that they can work also as coolants.
To maximize the heat dissipation is possible to eject a part of the propellant around the nozzle.

However, in space the major part of heat is dissipated as radiation.



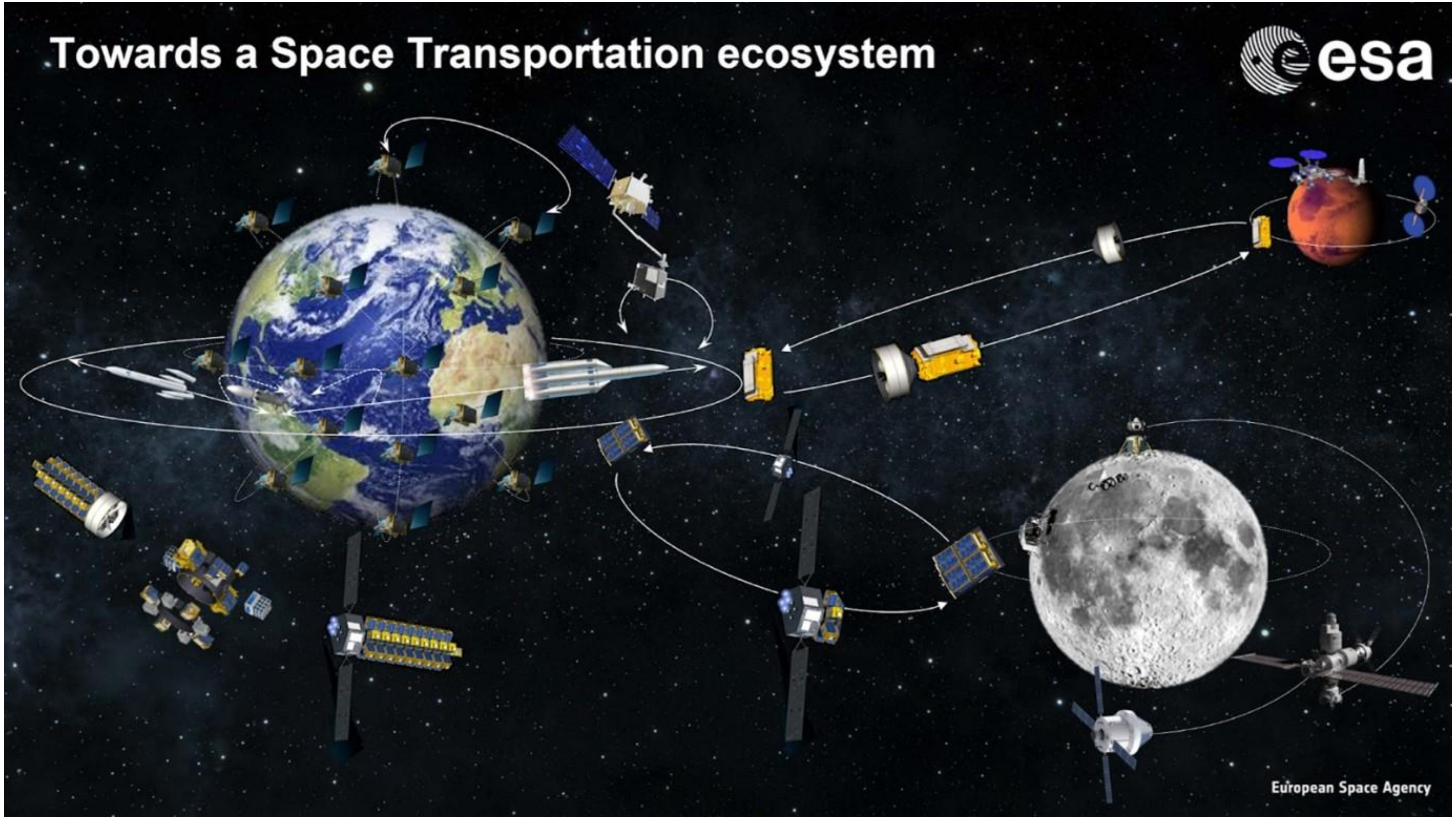
THE FUTURE OF SPACE LOGISTIC

What is space logistic?

Space logistics is "the theory and practice of driving space system design for operability and supportability, and of managing the flow of material, services, and information needed throughout a space system lifecycle."

It includes terrestrial logistics in support of space travel, movement of people in space, and contracting and supplying any required support services for maintaining space travel.

Space logistics has witnessed remarkable evolution over the years. From the early days of space exploration to the current era of satellite operations and space tourism, the logistics involved in accessing, maneuvering, and maintaining spacecraft in orbit have become increasingly sophisticated. As the demand for space-based activities grows, **space logistics is set to play a pivotal role in shaping the future of space exploration and satellite operations.**


Towards a Space Transportation ecosystem
esa
European Space Agency

ESA Space Transportation ecosystem

ESA studies indicate that a cost-effective and performant space transportation system on the 2025-2050 horizon could rely on an **optimised fleet of reusable launchers** injecting payloads on high parking orbits, combined with a “hub and spoke” **space logistics network to reach the final orbits** (e.g. constellations phasing, exploration missions…) and provide transportation support for in-orbit servicing.

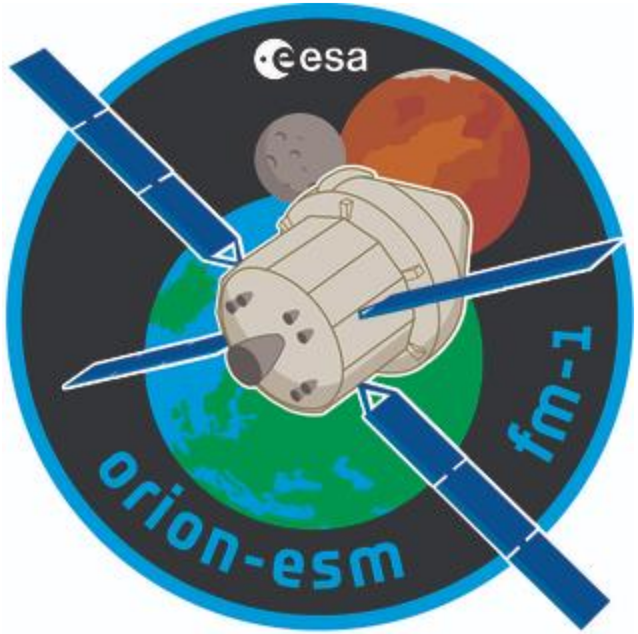
At the 2030 horizon, such a European ecosystem shall provide the new transportation services to- and in-space that will be required by the space users markets. This ecosystem shall provide logistics and transportation to, among others, the following use-cases and service providers:

★ **In-orbit servicing**: refuelling, maintenance, life-extension, storage, propellant depot
★ **In-space manufacturing and assembly**
★ **End-of-life management and active debris removal**
★ **In-orbit infrastructures for energy and/or data exchanges**
★ **Return to Earth**

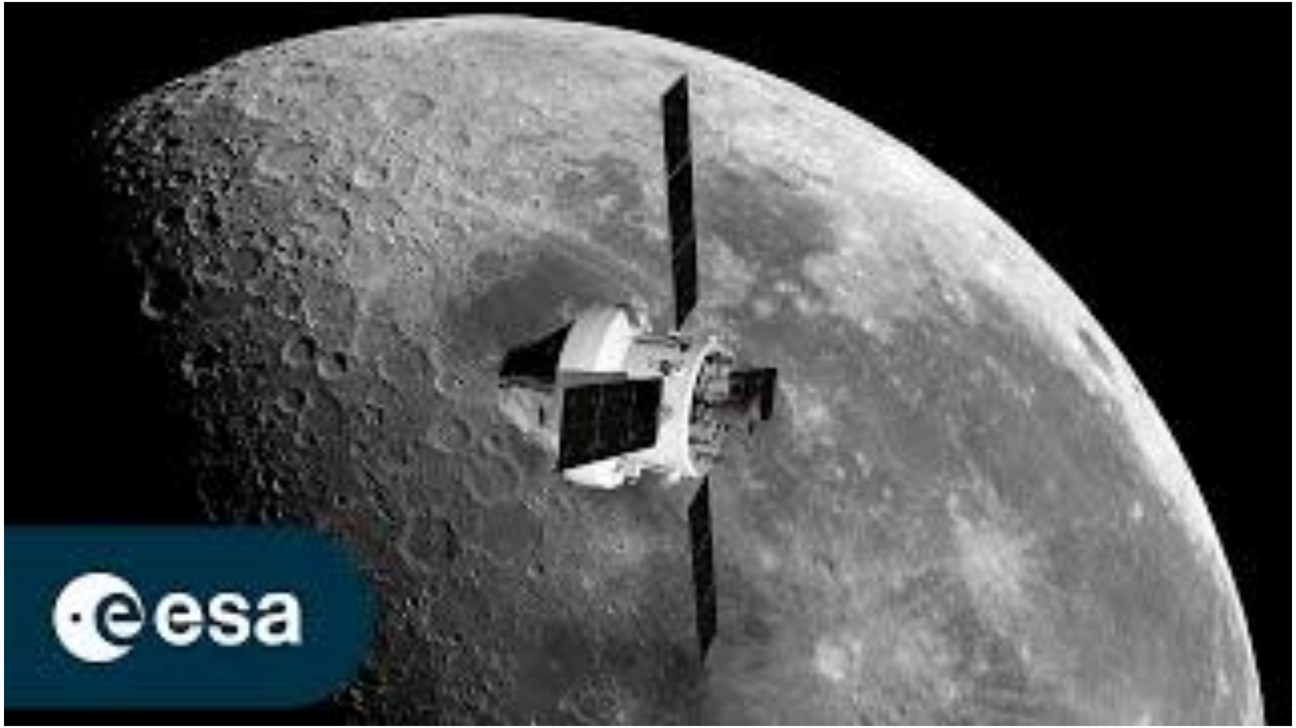
Transfer vehicles - European Service Module

The European Service Module (ESM) is the service module component of the Orion spacecraft, serving as its **primary power** and **propulsion component** until it is discarded at the end of each mission.

The service module provides in-space propulsion capability for orbital transfer, attitude control, and high altitude ascent aborts. It provides the **water** and **oxygen** needed for a habitable environment, **generates and stores electrical power**, and maintains the temperature of the vehicle's systems and components. This module can also transport **unpressurized cargo** and **scientific payloads**.



Artemis I – ESM perspective

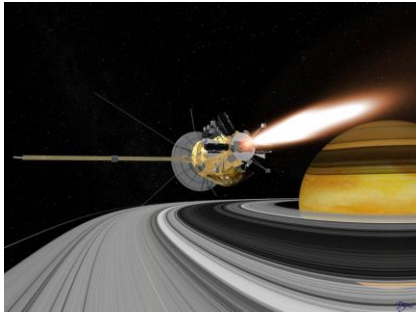


Interplanetary missions

Interplanetary missions are the crewed or uncrewed travel between stars and planets, usually within the Solar System. Their main goal is to gain scientific knowledge.

Hubble Telescope Picture

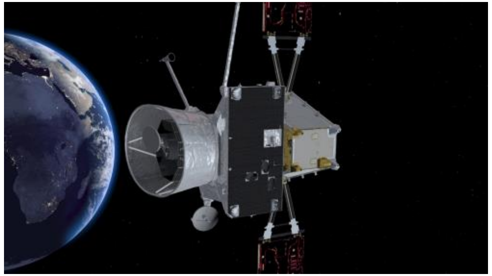
Interplanetary missions



Cassini Huygens

Voyager

Rosetta Probe



Bepi Colombo

What awaits us?

THANK YOU FOR YOUR ATTENTION